

SW. Engineering

LIBRARY MANAGEMENT SYSTEM

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Group 3
Section 5

SYSTEM DESCRIPTION.

This library management system (LMS) will make it easier for the university students to borrow books by automating the borrowing, returning and cataloging actions. It's designed for use by students, librarians and administrators. Providing a secure, simple, easy to use and efficient way for all users to benefit from it and manage the library resources.

1. REQUIREMENTS

1.1 Functional Requirements

- The system shall allow the user to look for the resources or textbooks they want via a search engine function.
- The system shall allow the user to borrow a selected book by inputting their full name, faculty ID, and the required borrowing duration.
- It should also provide a dashboard for librarians to catalog and organize the books based on the resources content such as subjects, categories or fields of study.
- The system shall notify the borrower of any fees assessed if a returned book or library resource is marked as damaged.

1.2. Non-Functional Requirements

- The system shall process user search, book selection, and borrowing requests with a response time not exceeding 3 seconds.
- The system shall securely store all user's passwords and private information using high standard encryption.
- The system shall guarantee an allowable downtime of maximum 30 minutes per day.
- The system shall be cross platform compatible, functioning on standard web browsers, Android devices, and major operating systems (Windows, macOS, and Linux).

2. USER STORIES

2.1 User Story (1)

2.1.1 card

- As a **student**, I want to **search for textbooks by subject**, so that I can **easily find the exact resources I need for my classes**.

2.1.1 Acceptance criteria

- The system should display a **search bar** on the home screen.
- The system should display the result of the search whether the book is available or not.

2.2 User Story (2)

2.2.1 card

- As a **librarian**, I want to **organize the textbooks based on field of study**, so that **resources are well organized and cataloged**.

2.2.2 Acceptance criteria

- The system should have "tagging" as a function to allow librarians put books under certain categories
- The system must provide a filtering option so users can browse and view all resources grouped under the same category.

2.3 User Story (3)

2.3.1 card

- As an **administrator**, I want to **set a fee assessed for damaged books**, so that **borrowers take better care of resources**.

2.3.2 Acceptance criteria

- The system should have an online payment system so users can pay cashless with their bank accounts
- The system must provide a settings page where the penalty fee amount can be inputted and updated.

3. TRACEABILITY

	Functional Requirement	Linked Non-Functional Requirement
1	Search for resources and borrow textbooks using student information.	NFR 1, NFR 2, NFR 3, NFR 4
2	Catalog and organize library resources based on their subject via a librarian dashboard.	NFR 2, NFR 3, NFR 4
3	Assess penalty fees for damaged books and process secure online payments.	NFR 2, NFR 3, NFR 4

4. MAIN CLASSES

4.1 Student

4.1.1 purpose

Representing the person borrowing the book.

4.1.1 attributes

- Name: String
- FacultyID: int
- Email: String
- BorrowedBooksList: List<Book>

4.2 Book

4.2.1 purpose

Representing the resource / item being borrowed

4.2.2 attributes

- Title: String
- Author: String
- Subject: String
- isAvaible: Boolean
- isDamaged: Boolean

4.3 Librarian

4.3.1 purpose

Representing the library keeper and organizer

4.3.2 attributes

- Name: String
- EmployeeID: int
- WorkingHours: int
- Email: String

4.4 Borrows

4.4.1 purpose

Representing the event of borrowing a book by a student

4.4.2 attributes

- Book: Book
- Borrower: Student
- BorrowDate: Date
- ReturnDate: Date

4.5 Fine

4.5.1 purpose

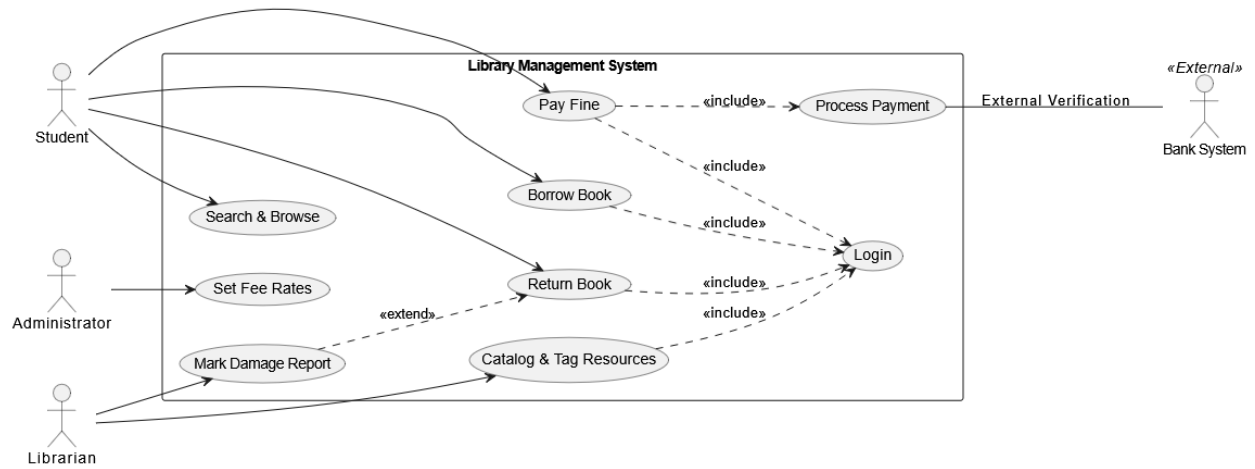
Representing the penalty fee required of student damaged a borrowed resource

4.5.2 attributes

- Price: Double
- Reason: String
- isPaid: Boolean
- dmgCount: int (representing the number of times a student returned a damaged resource)

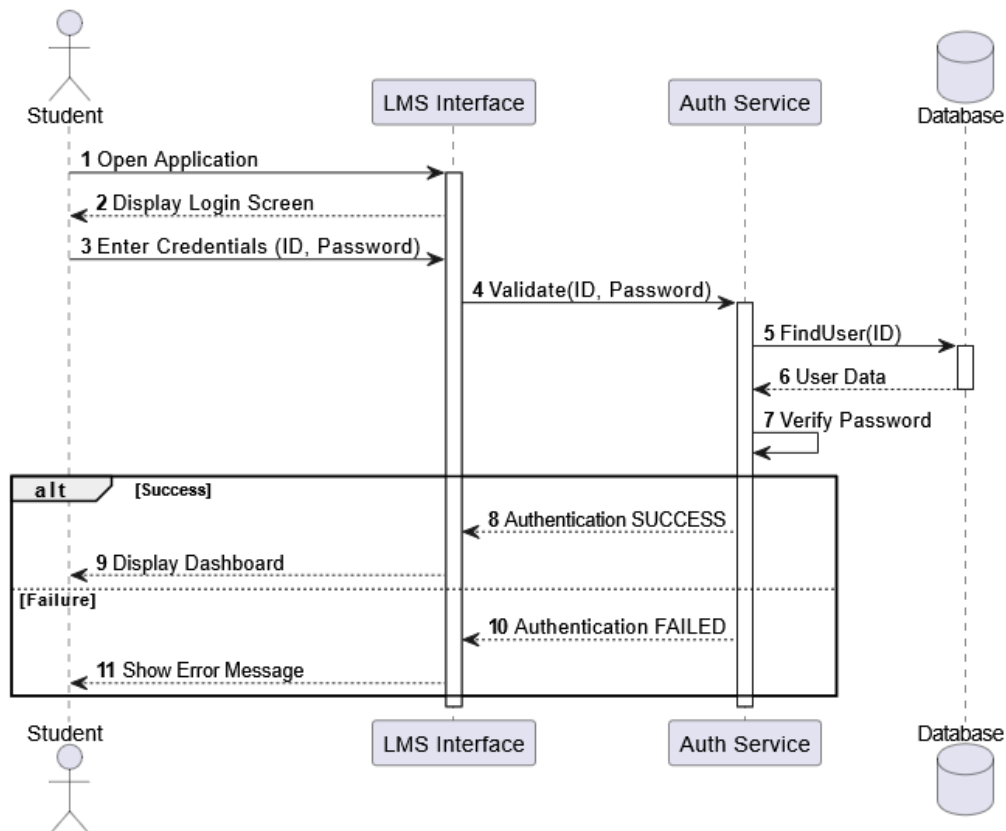
5. DIAGRAMS

5.1 Use Case Diagram

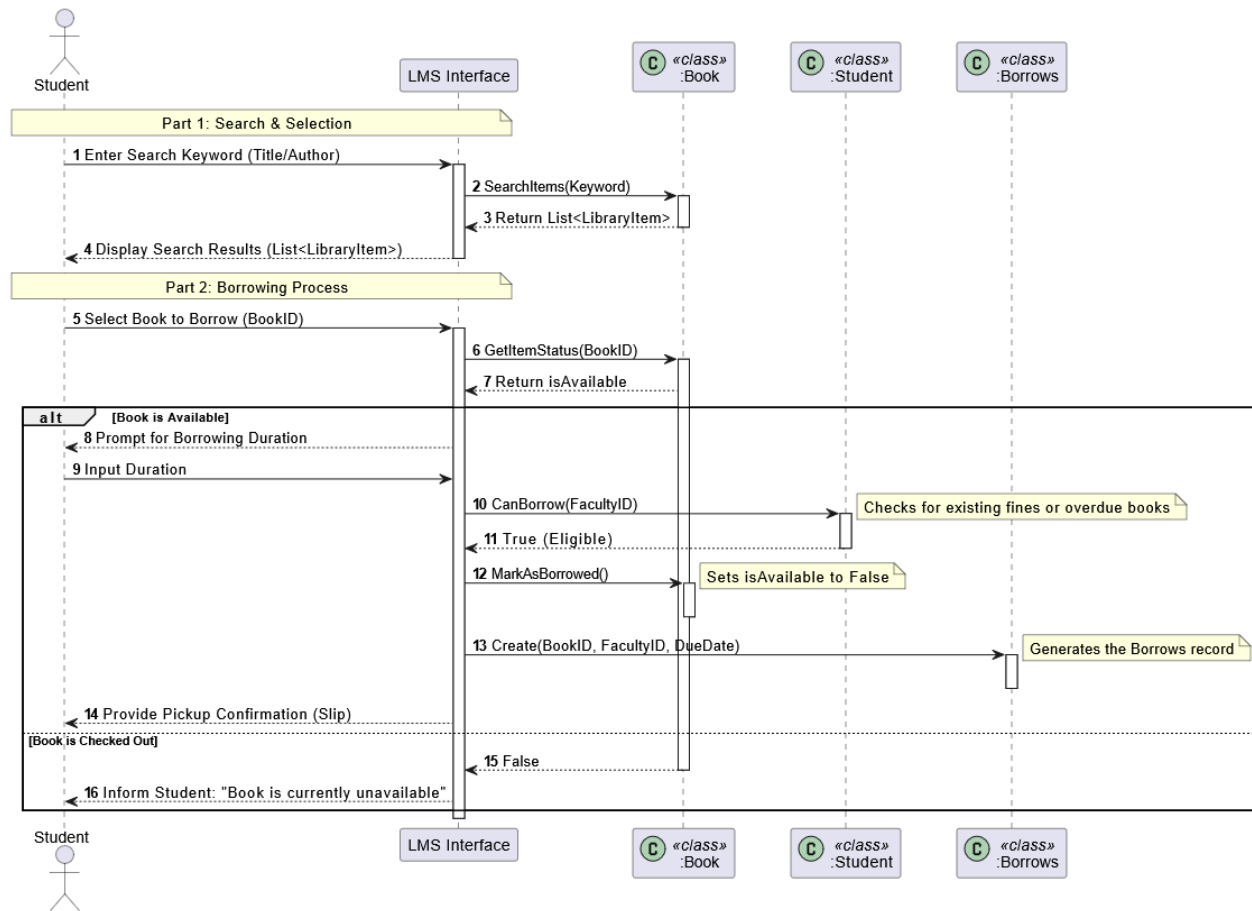


5.2 Sequence Diagrams

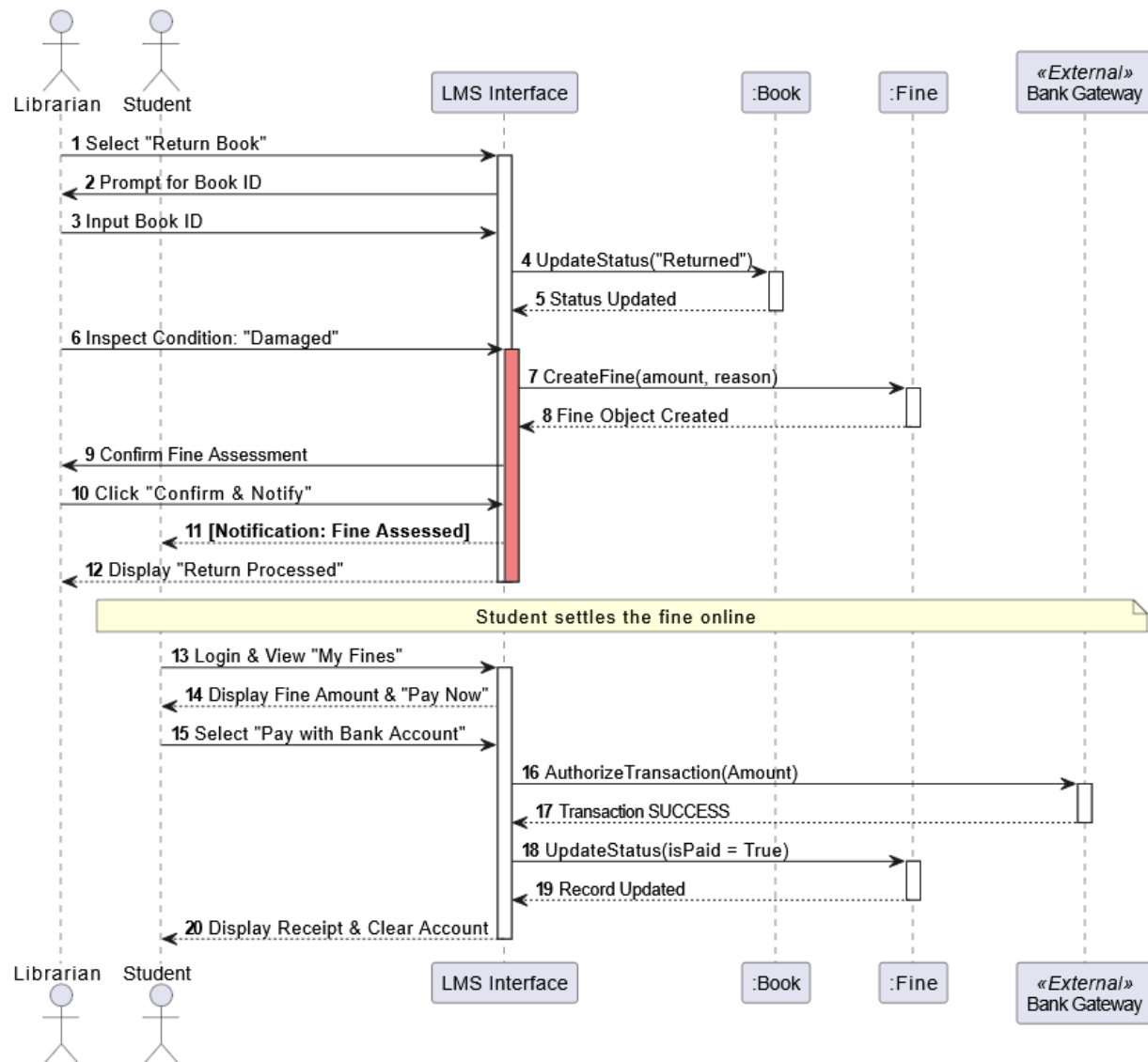
Log-in sequence



Borrowing a book sequence

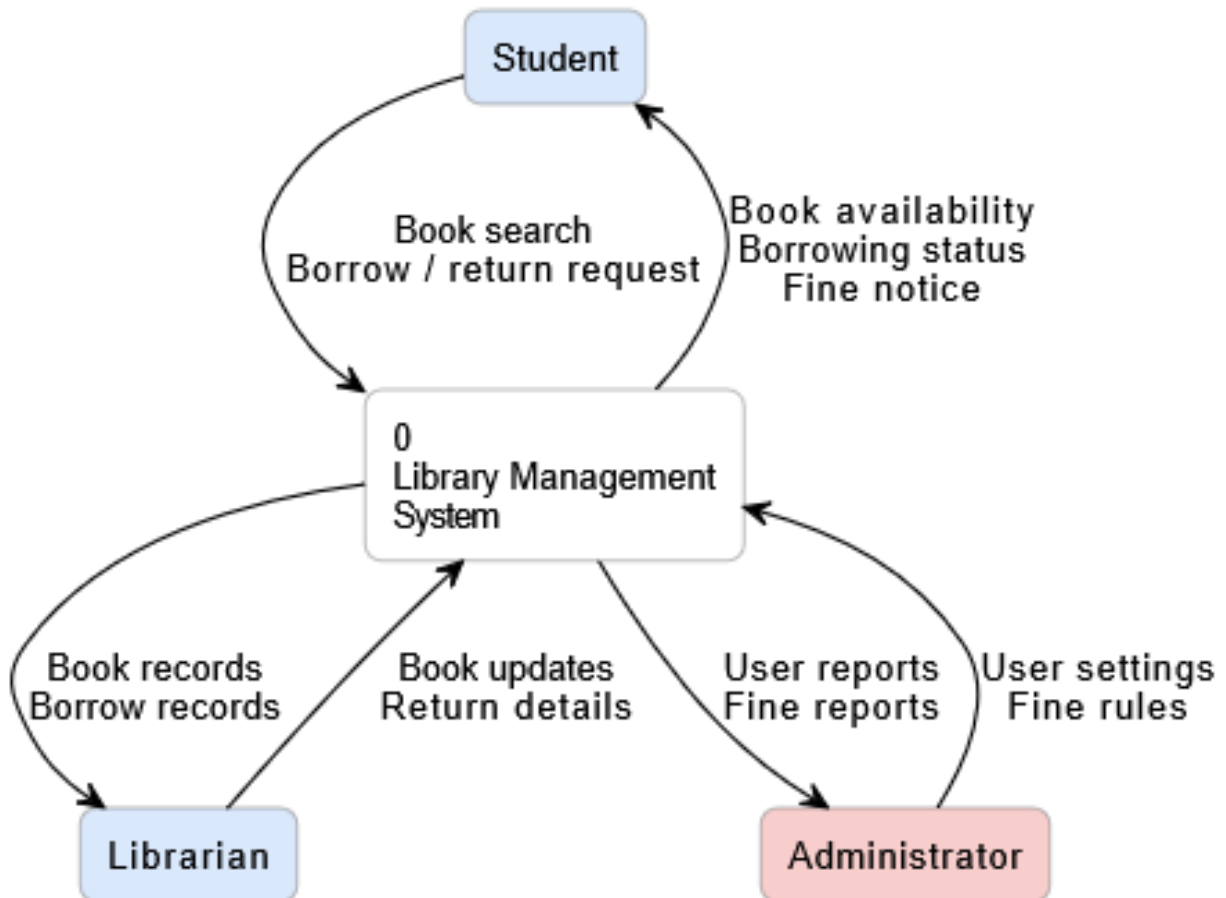


Marking a return as damaged and student pays the fee sequence

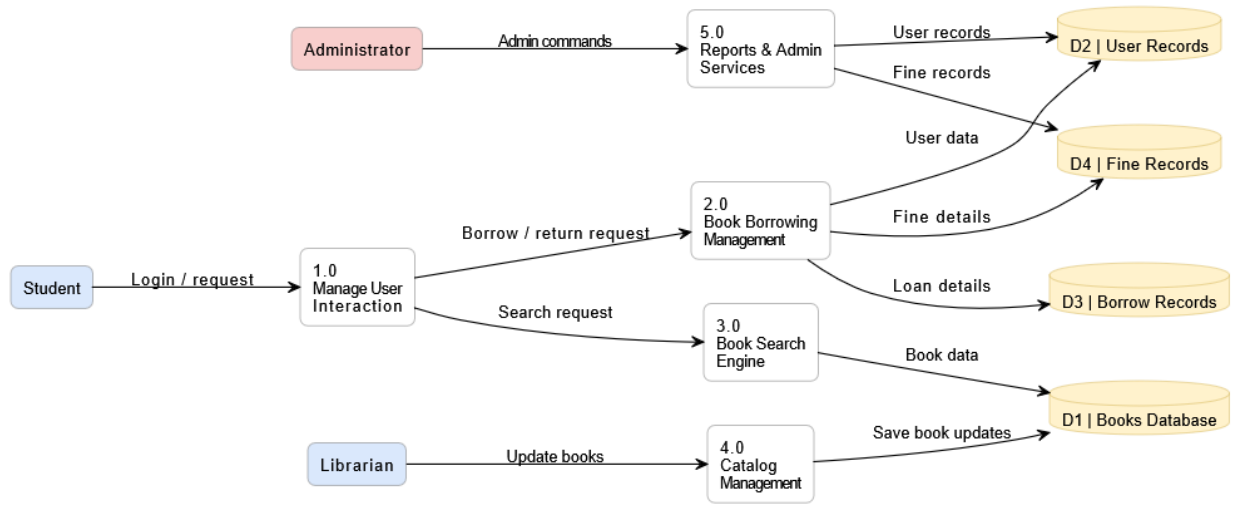


5.3 DFD diagrams

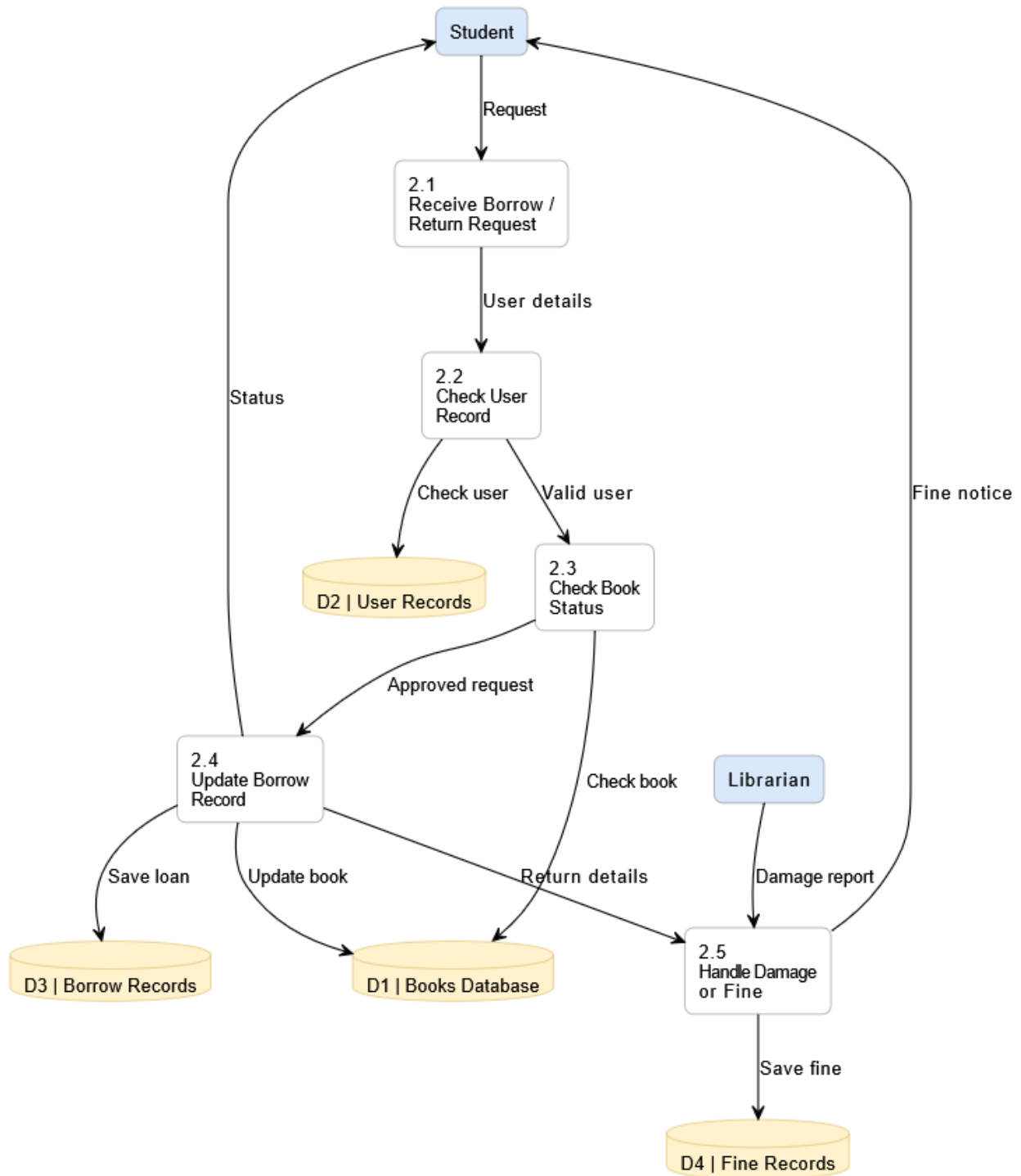
DFD 0-Level / Context Diagram - Library Management System



DFD 1-Level - Library Management System



DFD 2-Level - Book Borrowing Management



6. CYCLOMATIC COMPLEXITY

There are **2 decision statements**:

1. IF username == "admin" AND password == "admin123"
2. ELSE IF username == "user" AND password == "user123"

$V(G) = \text{Number of decisions} + 1$

$V(G) = 2 + 1 = 3$

Cyclomatic Complexity = 3

Three independent paths:

Path1: Admin login successful

Start

Display "Enter Username"

Input username

Display "Enter Password"

Input password

Check admin condition = TRUE

Display "Welcome, Administrator."

Call LoadAdminDashboard

End

Path2: User login successful

Start

Display "Enter Username"

Input username

Display "Enter Password"

Input password

Check admin condition = FALSE

Check user condition = TRUE

Display "Welcome, User."

Call LoadUserDashboard

End

Path3: Invalid credentials

Start

Display "Enter Username"

Input username

Display "Enter Password"

Input password

Check admin condition = FALSE

Check user condition = FALSE

Display "Error: Invalid Credentials."

Call ShowLoginScreen

End
